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Patient-Reported Outcomes

Translation and Testing Psychometric Properties of the Thai Version of Vision Bolt-On EQ-5D in Patients With Visual Impairment

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Highlights

- We addressed the lack of validated Thai versions of the vision bolt-on EQ-5D by translating, culturally adapting, and psychometrically testing these instruments in a sample with varying degrees of visual impairment.
- Our findings revealed moderate to good reliability, strong known-group validity, and moderate responsiveness for both the 3L and 5L versions, with the 5L demonstrating higher sensitivity and reduced ceiling effects.
- Incorporating a vision bolt-on dimension in Thai version of the EQ-5D can guide evidence-based clinical decisions and economic evaluations, potentially improving outcomes for individuals with visual impairments.

Abstract

Objectives

This study aimed to translate, culturally adapt, and evaluate the psychometric properties of the Thai versions of the vision bolt-on EQ-5D-3L and EQ-5D-5L, comparing them with the standard EQ-5D-3L, EQ-5D-5L, and EQ-VAS in individuals with varying degrees of visual impairment.

Methods

Following EuroQol Translation Guidelines, the vision bolt-on items were translated through forward-backward processes and underwent cognitive debriefing. A total of 301 participants were interviewed at baseline, with 275 completing a 2-week follow-up. Known-groups validity was assessed based on visual acuity classifications and

participants who reported better or worse vision.

Results

Both versions of the vision bolt-on had no missing data and demonstrated moderate to good test-retest reliability (ICC range 0.62-0.64). They effectively differentiated quality-of-life scores across varying degrees of vision impairment, showing reduced ceiling effects (5%-9%) compared with the standard EQ-5D. Correlations with EQ-VAS were moderate (rho 0.40-0.45). When participants reported improved vision, the vision bolt-on EQ-5D-5L exhibited the highest responsiveness, whereas all instruments were generally less responsive for worsening vision.

Conclusions

The Thai versions of the vision bolt-on EQ-5D-3L and EQ-5D-5L demonstrated acceptable reliability, validity, and moderate responsiveness for capturing vision-related health changes, with the 5L version showing slightly greater sensitivity. These findings support the usefulness of adding a vision dimension to EQ-5D for Thai-speaking populations, facilitating more precise evaluations of interventions targeting visual health.

Keywords

EQ-5D; EuroQoL; ophthalmology; quality of life; visual impairment

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Introduction

Health-related quality of life (HRQoL) measurement is pivotal in evaluating healthcare interventions, informing policy decisions, and understanding patient outcomes in chronic and disabling conditions. The EQ-5D, developed by the EuroQol Group, is a widely used generic instrument for assessing HRQoL because of its brevity, cross-cultural applicability, and utility in economic evaluations.[1] The original EQ-5D-3L and its expanded counterpart, the EQ-5D-5L, which offers improved sensitivity with 5 response levels per dimension, have been validated across diverse populations.[2,3] However, the standard EQ-5D's 5 core dimensions—mobility, self-care, usual activities, pain/discomfort, and anxiety/depression—may inadequately capture the burden of specific health issues, such as visual impairment, which profoundly impacts daily functioning and psychosocial well-being.[4] This limitation is highlighted in a recent multicenter Thai study on patients with eye diseases, which found that the standard EQ-5D-5L showed a significant response only when visual impairment became severe, failing to detect the impact of mild or moderate impairment. Furthermore, the same study found that the Health Utility Index 3, an instrument that includes a vision domain, demonstrated a stronger correlation with a vision-specific quality of life tool than the standard EQ-5D-5L did.[5]

discriminative power and utility measurement of the EQ-5D in various contexts.[6] Vision-related bolt-ons, for instance, aim to quantify the unique challenges faced by individuals with visual impairment.[7] This additional item asks respondents to rate their vision-related difficulties using the same 3- or 5-level response structure as the core EQ-5D dimensions and has been proven to increase responsiveness and discriminatory power of the questionnaire.[8,9]

In Thailand, visual impairment continues to pose a significant public health challenge, affecting >2% of the population. Conditions such as cataracts, diabetic retinopathy, and glaucoma notably reduce individuals' quality of life (QoL).[10] Despite the availability of the Thai EQ-5D-3L and EQ-5D-5L,[11] no validated vision-specific bolt-on exists for these tools. Cross-cultural adaptation requires meticulous translation, cognitive debriefing, and psychometric validation to ensure linguistic and conceptual equivalence.[12,13] Furthermore, comparing the performance of the 3L and 5L versions with a vision bolt-on could clarify whether the enhanced granularity of the 5L improves sensitivity in detecting vision-related QoL changes. To our knowledge, no psychometric evaluation has been conducted on non-English versions of vision-related QoL questionnaires.

This study aims to translate and evaluate the psychometric properties of Thai versions of the vision bolt-on EQ-5D-3L and EQ-5D-5L questionnaires and to compare the properties with the original translated EQ-5D questionnaires. Specifically, we assessed their reliability (test-retest stability), constructed validity (convergent and known-groups), and responsiveness in a cohort of Thai adults with varying degrees of visual impairment. By addressing this gap, our findings will inform the use of these adapted tools in clinical practice, health economic evaluations, and population studies targeting vision-related QoL in Thailand.

Methods

This prospective observational study was approved by the Institutional Review Board of the Faculty of Medicine, Chulalongkorn University (COA No. 1023/2023). The study was conducted in accordance with the Declaration of Helsinki. All participants received a comprehensive explanation of the study, and informed consent was obtained before enrollment.

Translation of the Vision Bolt-On EQ-5D-3L and the Vision Bolt-On EQ-5D-5L

Permission for use and translation of the vision bolt-on EQ-5D-3L and the vision bolt-on EQ-5D-5L was granted without charge by the EuroQol Research Foundation. The translation process followed the required EuroQol Translation Guidelines.[14] The process consists of forward translation, backward translation, cognitive debriefing, and submission for approval from the EuroQol. Two independent bilingual experts carried out the forward translation of the vision bolt-on component, referencing the legacy Thai EQ-5D-3L and EQ-5D-5L versions. Subsequently, the translators and investigators conferred to produce a consensus version. The backward translation was done by 2 independent translators, and the reconciled version was compared to the original English version of the questionnaires to produce the second consensus

cognitive debriefing process, and it was then proofread by another native linguist. The consensus versions of every process were approved by the EuroQol.

Patients and Procedures

A sample size was determined using G*Power version 3.1.9.7 (Heinrich-Heine-Universität Düsseldorf, Germany),[15] assuming a moderate effect size for test-retest reliability, known-groups validity, convergent validity, and responsiveness, with 90% power and a 0.05 significance level. The largest calculated sample size based on known-groups validity and a projected 30% attrition rate was 300. The participant was conveniently sampled from the Ophthalmology Clinic at the King Chulalongkorn Memorial Hospital from February 1 to March 31, 2024. Participants were categorized into 5 groups on the basis of the WHO International Classification of Diseases 11 (ICD-11) criteria for vision impairment.[16] Those with no vision impairment had a best-corrected visual acuity (BCVA) of at least 20/40 in the better eye. Mild vision impairment was defined as a BCVA worse than 20/40 but equal to or better than 20/70, while moderate vision impairment included individuals with a BCVA worse than 20/70 but equal to or better than 20/200. Severe vision impairment was classified as a BCVA worse than 20/200 but equal to or better than 20/400. Finally, blindness was defined as a BCVA worse than 20/400 in the better eye.

Participants were asked to complete the translated questionnaires in the following sequence: (1) EQ-5D-5L; (2) vision bolt-on EQ-5D-5L; (3) participants' demographic data, including age, sex, educational level, and past ophthalmic and systemic diseases; (4) EQ–visual analog scale (VAS); (5) EQ-5D-3L; and (6) vision bolt-on EQ-5D-3L. If participants were unable to read the questionnaires, interviewers would read them to the participants without any interpretation. Two weeks after the first interview, participants would be contacted by the same interviewer by means of a video call. Participants were asked to compare their current eye health perception with their initial assessment using the following options: (1) much better than the first assessment, (2) somewhat better than the first assessment, (3) about the same as the first assessment, (4) somewhat worse than the first assessment, and (5) much worse than the first assessment. Patients who reported no change in eye health[3] were included in the test-retest analysis. Those who indicated an improvement[1,2] or a decline in eye health[4,5] were categorized into the improvement and worsening groups, respectively, for the responsiveness assessment.

The utility scores of the EQ-5D-3L and EQ-5D-5L were calculated using Thai value sets,[17,18] whereas the utility scores of the vision bolt-on EQ-5D-3L and vision bolt-on EQ-5D-5L were calculated using available value sets from other countries.[7,19]

Data Analyses

Practicality was assessed based on the time required to complete the questionnaire, recorded by the interviewers. The ceiling effect was analyzed on the basis of the percentage of participants who achieved the highest score on the questionnaire. If this percentage was <15%, the questionnaire was considered capable of distinguishing between individuals with different health conditions.[20] Test-retest reliability was evaluated using intraclass correlation coefficients (ICCs), calculated with the two-way mixed and single-measure method. ICC values were interpreted as follows: <0.40

correlation coefficients was as follows: values below 0.40 indicated poor agreement, those between 0.40 and 0.75 represented fair to good agreement, and values above 0.75 reflected excellent agreement.[22]

Known-groups validity was assessed by evaluating differences in QoL between groups with and without the characteristic of interest. Statistical analysis was performed using the Mann-Whitney U or Kruskal-Wallis test. For the Mann-Whitney U test, effect sizes were categorized as small (<0.3), medium (0.3-0.5), and large (>0.5).[23] For the Kruskal-Wallis test, effect sizes were classified as small (0.01-0.059), medium (0.06-0.13), and large (>0.13).[23] Convergent validity was evaluated by comparing the relationship between instruments measuring similar health concepts using Spearman's rank correlation coefficients. The strength of the correlation was determined using established criteria as follows: weak or no correlation ($r < 0.25$), moderate ($0.25 \le r < 0.50$), moderate to strong ($0.50 \le r < 0.75$), and strong ($r \ge 0.75$).[24] Responsiveness refers to the ability of the questionnaires to detect changes in eye health status over time. This was assessed using scores from patients who perceived their eye health status as improved or worsened at follow-up. Responsiveness was evaluated by calculating the standardized effect size to compare QoL scores between the 2 time points. A value of 0.2 indicated low sensitivity to eye health change, 0.5 represented moderate sensitivity, and 0.8 suggested high sensitivity to eye health change.[24] All statistical analyses were done using IBM SPSS version 29. $P < .050$ were considered statistically significant.

Results

The study initially recruited 421 participants, but 120 individuals declined to participate, resulting in a refusal rate of 28.5%. This reduced the number of participants to 301 for the first interview. Approximately 46% of participants self-completed the questionnaires, whereas the remaining responses were read aloud by researchers without any interpretation. Following the first interview, 26 participants were unable to complete the second interview, with 3 declining to be interviewed and 23 being unreachable. Ultimately, 275 participants completed the second interview. Baseline characteristics of the participants and participant flow chart are presented in Appendix 1 in Supplemental Materials found at https://doi.org/10.1016/j.vhri.2025.101498 ↗. Most of the participants were women and were between 61 and 70 years old. Around 53% of participants had a bachelor's degree or higher. Retinal diseases were the most common concurrent ocular disease among participants, and >80% of participants had no vision impairment with the better-seeing eye. Most participants (78.1%) had at least 1 systemic comorbidity. EQ-5D-5L and its vision bolt-on variant showed higher mean scores than their EQ-5D-3L counterparts (Table 1).

Table 1. Questionnaire scores, response time, and ceiling effects.

Questionnaires	Mean score (SD)	Median score (IQR)	Score range	Response time (mean (SD)
EQ-5D-3L	0.75 (0.21)	0.71 (0.61-1.00)	−0.15 to 1.00	1:06 (0:36)
EQ-5D-5L	0.87 (0.13)	0.89 (0.82-0.94)	0.17-1.00	1:33 (0:50)

Vision bolt-on EQ-5D-3L	0.91 (0.10)	0.93 (0.86-0.99)	0.16-1.00	0:18 (0:12)
Vision bolt-on EQ-5D-5L	0.81 (0.17)	0.84 (0.76-0.92)	−0.24 to 1.00	0:24 (0:16)

IQR indicates interquartile range; VAS, visual analogue scale.

The practicality assessment revealed that all questionnaires were fully completed, with no missing data or refusals after participants agreed to participate, demonstrating full compliance and engagement. The results indicate that the EQ-5D-5L is slightly more time-consuming but reduces ceiling effects compared to the EQ-5D-3L. The original EQ-5D questionnaires had higher ceiling effects. The vision bolt-on components add minimal time burden while maintaining low ceiling effects below 15% (9.3% and 5.32% for vision bolt-on EQ-5D-3L and EQ-5D-5L, respectively) (Table 1). The test-retest reliability, measured using ICC, indicated equally moderate to good agreement in all instruments (Table 2). Weighted kappa showed that the vision bolt-on dimension had good agreement in both versions, with a slightly higher kappa value in EQ-5D-5L (0.66) compared to EQ-5D-3L (0.59) (Table 3). The vision bolt-on had the highest kappa value among all domains in the EQ-5D-5L, whereas it ranked third in the EQ-5D-3L.

Table 2. Intraclass correlation of the questionnaires.

Questionnaires	**Single measures ICC (95% CI)**	***P* value**
EQ-5D-3L	0.63 (0.51-0.72)	<.001
EQ-5D-5L	0.65 (0.53-0.75)	<.001
EQ-VAS	0.58 (0.45-0.69)	<.001
Vision bolt-on EQ-5D-3L*	0.62 (0.48-0.73)	<.001
Vision bolt-on EQ-5D-5L*	0.64 (0.49-0.75)	<.001

Note. ICC of <0.40 indicates poor agreement; 0.40-0.74 indicates fair to good agreement; and ≥0.75 indicates excellent agreement.

ICC indicates intraclass correlation.

* The ICC was calculated on the overall utility score that combines the standard EQ-5D with the new vision bolt-on.

Table 3. Weighted kappa of each domain of the questionnaires.

Domains	**Weighted kappa (95% CI)***	
	EQ-5D-3L	**EQ-5D-5L**
Mobility	0.62 (0.47-0.76)	0.52 (0.38-0.65)
Self-care	0.70 (0.42-0.98)	0.62 (0.33-0.91)

Usual activities	0.44 (0.26-0.61)	0.42 (0.25-0.59)
Pain/discomfort	0.39 (0.23-0.54)	0.42 (0.29-0.56)
Anxiety/depression	0.57 (0.42-0.72)	0.47 (0.34-0.60)
Vision bolt-on	0.59 (0.46-0.73)	0.66 (0.55-0.76)

Note. Weighted kappa <0.40 indicates poor agreement; 0.40-0.75 indicates fair to good agreement; and >0.75 indicates excellent agreement.

*
All *P* values are < .001.

In the known-group validity assessment, no statistically significant differences in mean scores were observed across different sex and age groups in any of the questionnaires. Although significant differences were observed in education levels for EQ-5D-3L, EQ-5D-5L, and vision bolt-on EQ-5D-3L, the effect sizes were small. All questionnaires were sensitive to different numbers of comorbidities and the level of vision impairment in the better-seeing eye. The vision bolt-on domains demonstrated stronger effect sizes among the vision impairment categories, particularly when distinguishing on the basis of the visual acuity of the worse-seeing eye (Table 4). In terms of convergent validity, both EQ-5D-5L and vision bolt-on EQ-5D-5L exhibited similar correlations with EQ-VAS (Spearman's rho = 0.404 and 0.402, respectively), whereas the vision bolt-on EQ-5D-3L demonstrated a slightly stronger correlation with EQ-VAS compared to the EQ-5D-3L (Spearman's rho = 0.442 vs 0.405) (Table 5). All questionnaires showed moderate responsiveness to eye health status in the improvement group (Table 6). The instruments were generally less responsive to worsening health. The vision bolt-on EQ-5D-5L demonstrated the strongest responsiveness to both eye health improvements and deteriorations (standardized effect size = 0.486 and 0.659, respectively).

Table 4. Known-groups validity assessment of the questionnaires.

Patient characteristics	**EQ-5D-3L**	**EQ-5D-5L**	**EQ-VAS**	**Vision bolt-on EQ-5D**
Sex				
Male (N = 123)	0.77 (0.22)	0.88 (0.12)	0.75 (0.14)	0.91 (0.12)
Female (N = 178)	0.74 (0.20)	0.87 (0.13)	0.74 (0.15)	0.91 (0.09)
Effect size	0.08	0.05	0.04	0.04
P value	.190*	.390*	.500*	.460*
Age group				
18-40 years (N = 34)	0.77 (0.22)	0.87 (0.12)	0.77 (0.15)	0.90 (0.12)
41-50 years (N = 39)	0.75 (0.20)	0.88 (0.15)	0.78 (0.17)	0.90 (0.11)
51-60 years (N = 66)	0.76 (0.18)	0.90 (0.09)	0.71 (0.14)	0.91 (0.08)
61-70 years (N = 97)	0.76 (0.21)	0.88 (0.12)	0.74 (0.15)	0.91 (0.08)
>70 years (N = 65)	0.72 (0.24)	0.84 (0.16)	0.75 (0.15)	0.90 (0.17)

P value	.76†	.39†	.06†	.98†
Educational level				
Lower than bachelor's degree (N = 141)	0.73 (0.21)	0.86 (0.14)	0.73 (0.16)	0.90 (0.11)
Bachelor's degree or higher (N = 160)	0.77 (0.20)	0.89 (0.12)	0.76 (0.14)	0.92 (0.10)
Effect size	0.13	0.14	0.10	0.09
P value	**.02***	**.02***	.07*	.12*
Comorbidities				
None (N = 66)	0.86 (0.18)	0.93 (0.08)	0.81 (0.15)	0.94 (0.08)
1 (N = 95)	0.78 (0.19)	0.89 (0.10)	0.78 (0.12)	0.92 (0.08)
2 (N = 59)	0.73 (0.19)	0.88 (0.10)	0.73 (0.13)	0.90 (0.09)
≥3 (N = 81)	0.65 (0.22)	0.80 (0.17)	0.66 (0.16)	0.87 (0.13)
Effect size	0.11	0.11	0.14	0.09
P value	**<.001†**	**<.001†**	**<.001†**	**<.001†**
Vision impairment (better eye)				
No (N = 245)	0.78 (0.20)	0.89 (0.12)	0.76 (0.14)	0.92 (0.09)
Mild (N = 15)	0.65 (0.13)	0.84 (0.11)	0.71 (0.17)	0.91 (0.05)
Moderate (N = 28)	0.64 (0.24)	0.79 (0.18)	0.67 (0.18)	0.84 (0.17)
Severe (N = 1)	1.00 (NA)	1.00 (NA)	0.80 (NA)	0.99 (NA)
Blindness (N = 12)	0.59 (0.21)	0.80 (0.13)	0.66 (0.17)	0.78 (0.13)
Effect size	0.08	0.08	0.04	0.08
P value	**<.001†**	**<.001†**	**.01†**	**<.001†**
Vision impairment (worse eye)				
No (N = 139)	0.78 (0.19)	0.90 (0.11)	0.76 (0.13)	0.93 (0.07)
Mild (N = 12)	0.70 (0.14)	0.87 (0.08)	0.71 (0.16)	0.91 (0.06)
Moderate (N = 77)	0.77 (0.20)	0.87 (0.13)	0.74 (0.16)	0.91 (0.09)
Severe (N = 4)	0.53 (0.25)	0.68 (0.34)	0.64 (0.39)	0.77 (0.24)
Blindness (N = 69)	0.69 (0.24)	0.84 (0.14)	0.72 (0.16)	0.86 (0.14)
Effect size	0.04	0.04	0.02	0.07
P value	**.01†**	**.01†**	.33†	**<.001†**

Note. All data are presented as mean (SD); bold values indicate $P < .050$.

VAS indicates visual analogue scale.

*

Kruskal-Wallis tests: effect size of 0.01-0.059 indicates small; 0.06-0.13 indicates medium; and >0.13 indicates large.

Table 5. Spearman's rank correlation between questionnaires.

Spearman's rho (r)*	EQ-5D-5L	Vision bolt-on EQ-5D-5L	EQ-VAS	EQ-5D-3L	Vision bolt
EQ-5D-5L	1.000	0.872	0.404	0.828	0.721
Vision bolt-on EQ-5D-5L		1.000	0.402	0.728	0.793
EQ-VAS			1.000	0.405	0.442
EQ-5D-3L				1.000	0.843
Vision bolt-on EQ-5D-3L					1.000

Note. An r < 0.25 indicates weak or no correlation; ≥0.25 to <0.50 indicates moderate correlation; ≥0.50 to <0.75 indicates moderate to strong correlation; and ≥0.75 indicates strong correlation.

*
All *P* = .010.

Table 6. Responsiveness of the questionnaires.

Eye health status	Mean change	SD	*P* value*	SES
Better (N = 121)				
EQ-5D-3L	0.076	0.186	<.001	0.406
EQ-5D-5L	0.038	0.090	<.001	0.351
EQ-VAS	0.065	0.132	<.001	0.457
Vision bolt-on EQ-5D-3L	0.031	0.064	<.001	0.405
Vision bolt-on EQ-5D-5L	0.063	0.112	<.001	0.486
Worse (N = 24)				
EQ-5D-3L	−0.070	0.170	.06	0.267
EQ-5D-5L	−0.003	0.098	.82	0.014
EQ-VAS	−0.011	0.120	.67	0.052
Vision bolt-on EQ-5D-3L	−0.032	0.098	.15	0.158
Vision bolt-on EQ-5D-5L	−0.022	0.136	.91	0.659

Note. SES of 0.2 indicates low sensitivity; 0.5 indicates moderate sensitivity; and 0.8 indicates high sensitivity.

SES indicates standardized effect size.

*

Discussion

This study investigated the psychometric properties of the Thai versions of the vision bolt-on EQ-5D-3L and vision bolt-on EQ-5D-5L, comparing them with the standard EQ-5D-3L, EQ-5D-5L, and EQ-VAS. Overall, our findings suggest that adding the vision dimension offers enhanced sensitivity to vision-related health states. This is consistent with previous work on vision bolt-ons in other settings[8,9] and aligns with a broader research strategy of using task-specific bolt-ons for particular populations, such as adding a "respiratory" dimension for patients with chronic obstructive pulmonary disease[25] or "interpersonal relationships" for the Thai population.[26] Below, we discuss our key results in the context of previous research on EQ-5D instruments and their adaptations.

In terms of practicality, our study demonstrated no missing data and a 0% refusal rate among participants who agreed to take part, indicating that the questionnaires were acceptable and feasible in this Thai population. The short completion times for the bolt-on items (a mean of 18-24 seconds) also support their practicality. These findings align with the general usability of the EQ-5D, which has been widely reported as brief and user-friendly.[2] Both instruments showed moderate to good test-retest reliability, with ICC values ranging from 0.62 to 0.64 and weighted kappa values from 0.59 to 0.66, which were similar to their original counterparts. The ICC values in this study aligned with the results from other psychometric studies of the EQ-5D conducted in similar Thai populations.[11,27] When placed in this context, these values are shown to be consistent and acceptable; for instance, a study by Pattanaphesaj et al[27] in Thai diabetes patients found a nearly identical ICC of 0.64 for the EQ-5D-3L index. Although some guidelines recommend ICC values above 0.70 for excellent reliability,[28] questionnaires capturing broad aspects of health with less number of items might yield lower ICC values.[29] Our results are also consistent with similar validation studies of the EQ-5D-5L, which also reported ICCs in the moderate-to-good range.[30] Notably, the vision bolt-on EQ-5D-5L showed slightly higher ICC values than the 3L version, suggesting that the additional scales may improve test-retest stability in capturing changes or consistency in vision-related health.

In line with our expectations regarding known-groups validity, participants with comorbidities and more severe vision impairment reported significantly lower QoL scores, indicating that both standard EQ-5D and vision bolt-on versions can differentiate between health states. These findings align with previous research indicating that generic instruments effectively capture major health decrements,[7] whereas the bolt-on additions can further distinguish the severity of visual impairment, as previously demonstrated in the original English version.[9] However, there were no significant differences by sex or age, possibly reflecting the cross-sectional nature of our data and the relatively narrow age distribution of participants. A notable finding was the significant differences by educational level (albeit with small effect sizes), suggesting that social determinants of health can influence self-reported QoL.[31] Correlations between EQ-5D versions and their vision bolt-ons were high (Spearman's rho 0.84-0.87), indicating that the bolt-on dimension is well integrated into the overall health construct with good convergent validity. Correlations with EQ-VAS were moderate (0.40-0.45), which aligns with previous studies noting that a single, overall self-rating (the VAS) may not fully capture the

specific issues contribute substantially to individuals' overall health perception.

Concerning responsiveness, when participants reported feeling "better," the vision bolt-on EQ-5D-5L showed the largest standardized effect size, indicating moderate sensitivity to improvements in health. This aligns with previous research demonstrating that the vision bolt-on dimensions can enhance the sensitivity of the EQ-5D to the benefit of cataract surgery.[8] However, for those who felt "worse," changes were generally not significant, which might reflect the relatively small number of participants in the "worsening" group or the possibility that subtle declines in eye health may be harder to detect in a short follow-up period. This asymmetrical responsiveness, in which health outcome instruments in ophthalmology can often detect improvement more robustly than they can detect decline, aligns with findings in other research.[33,34] Future longitudinal studies with larger samples experiencing deterioration could provide additional evidence on downward responsiveness. Our study also revealed that EQ-5D-3L exhibited higher ceiling effects (35%) than the EQ-5D-5L (23%)—a pattern consistently reported in the literature.[2,35] The vision bolt-on versions further reduced ceiling effects (5%-9%), which suggests that adding a vision dimension can help discriminate mild or moderate health problems that might otherwise go unnoticed on the standard instrument. This finding highlights the benefit of a bolt-on domain where specific aspects of functioning, such as vision, are critically relevant.

Our work correlates with the previous research by Gandhi et al[8] and Luo et al,[9] which demonstrated that including a vision dimension could enhance both the discriminative ability and responsiveness of EQ-5D in populations with eye conditions. Similarly, Herdman et al[2] and Janssen et al[35] reported that the EQ-5D-5L version generally improves measurement precision and reduces ceiling effects compared to the 3L. We extend these findings to a Thai-speaking population, confirming that the vision bolt-on is feasible and reliable in a non-English setting—a gap previously identified in the literature. The bolt-ons also enhance sensitivity without significantly increasing the response time of the questionnaires. Our results suggest that the Thai vision bolt-on can capture meaningful differences in vision-related QoL while retaining the advantages of a generic, preference-based measure.

This study has several notable strengths. First, it employed robust translation and validation procedures, including forward-backward translation, and a cognitive debriefing process, ensuring linguistic and conceptual equivalence of the Thai vision bolt-on versions. Second, the large sample size and full data completion rate underscore the feasibility and acceptability of these instruments for Thai participants. Third, including both 3L and 5L versions alongside the vision bolt-on items offers comprehensive insights into how each variant performs in terms of reliability, validity, responsiveness, and practicality. However, this study has several notable limitations. First, the sample was recruited via convenience sampling from a single tertiary hospital, resulting in a highly educated cohort that was also predominantly composed of individuals with no vision impairment in their better-seeing eye (over 80%). Consequently, the very small subgroups for severe impairment (N = 1) and blindness (N = 12) limit the generalizability of our findings to more severely impaired populations and warrant further validation in such cohorts. Second, the relatively small number of participants reporting worsening health

generalizability of these findings. Another important limitation is the absence of a Thai value set for the vision bolt-on utility score, which restricts the ability to derive preference-based utility estimates tailored to Thai cultural and societal contexts. Finally, the cross-sectional nature of most comparisons restricts conclusions about long-term changes in health status. Future longitudinal studies could help confirm the long-term responsiveness, predictive validity, and utility valuation of these instruments.

Conclusions

In conclusion, the Thai versions of the vision bolt-on EQ-5D-3L and EQ-5D-5L demonstrated acceptable reliability, strong known-group validity, and moderate responsiveness, particularly for detecting improvements in vision-related health. The instrument's responsiveness to worsening vision could not be confirmed, likely because of the small sample size of this subgroup. However, the 5L version seemed more sensitive overall, reducing ceiling effects to a greater extent and showing a higher effect size when participants reported feeling better, suggesting it may be more suitable for capturing subtler differences in vision-related QoL. These findings underscore the practical and clinical value of adding a vision dimension to EQ-5D instruments in Thai-speaking populations, enabling more precise evaluations of interventions aimed at visual health. Future longitudinal research is recommended to further confirm its performance in detecting health declines and to broaden its use.

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